

Sainfoin Cloning

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SOP STATUS: Completed ▾ **REVIEW STATUS:** Approved ▾

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PURPOSE

The purpose of this document is to detail the main procedures and materials needed to take cuttings from a mature sainfoin plant from the field or greenhouse, and propagate them, taking care to minimize disease, etc. The purpose of cloning sainfoin plants is to maintain original genotypes for use in synthetic populations/cultivars and replicated/multisite genomic studies since sainfoin seeds do not produce plants which are true-to-type.

SCOPE

The scope of this document is limited to asexually producing clonal explants from a rooted, mature sainfoin plant in the greenhouse. It does not include any other type of asexual propagation method such as tissue culture, etc. Furthermore this document is limited to the current best practices for sainfoin cloning/cutting propagation used at The Land Institute and ATU (Adana, Turkey) and does not include methodologies or materials currently being evaluated elsewhere to improve the procedure.

REFERENCES AND RELATED DOCUMENTS

Avci, S., Cocu, S., Aasim, M., Sancak, C., & Ozcan, S. (2010). Effects of treating with auxin solutions on rooting of cuttings of sainfoin (*Onobrychis viciifolia*). *Tropical Grasslands*, 44, 123-127.

MATERIALS

- Scissors and Sharp razor blade
- Rooting Hormone with IBA Indole-3-butyric-acid (i.e. [Clonex Rooting Gel](#))
- Optional: Cloning solution with IBA Indole-3-butyric-acid (i.e. [Clonex Cloning Solution](#))
- Small vial to dip clones in gel
- Net pots or standard molded pots (approx 20mL in volume)
- Greenhouse pot trays (picture 1)
- Perlite (picture 2)
- Large pot or container to hold perlite-water mixture (picture 3 and 4)
- Spray bottle
- [Zerotol 2.0](#) concentrate solution diluted with water to 2% (picture 5)

Picture 1



Picture 2



Picture 3



Picture 4



Picture 4



Figure 1. Materials needed for cloning sainfoin plants.

DEFINITIONS

- Elongated stem: Stems with internode elongation and leaves at each node. Flower buds may begin to develop (Figure 2.a)
- Crown Bud: A crown bud is located at the base or crown of the plant just above the soil, where the main stem meets the root system. (Figure 2.b)

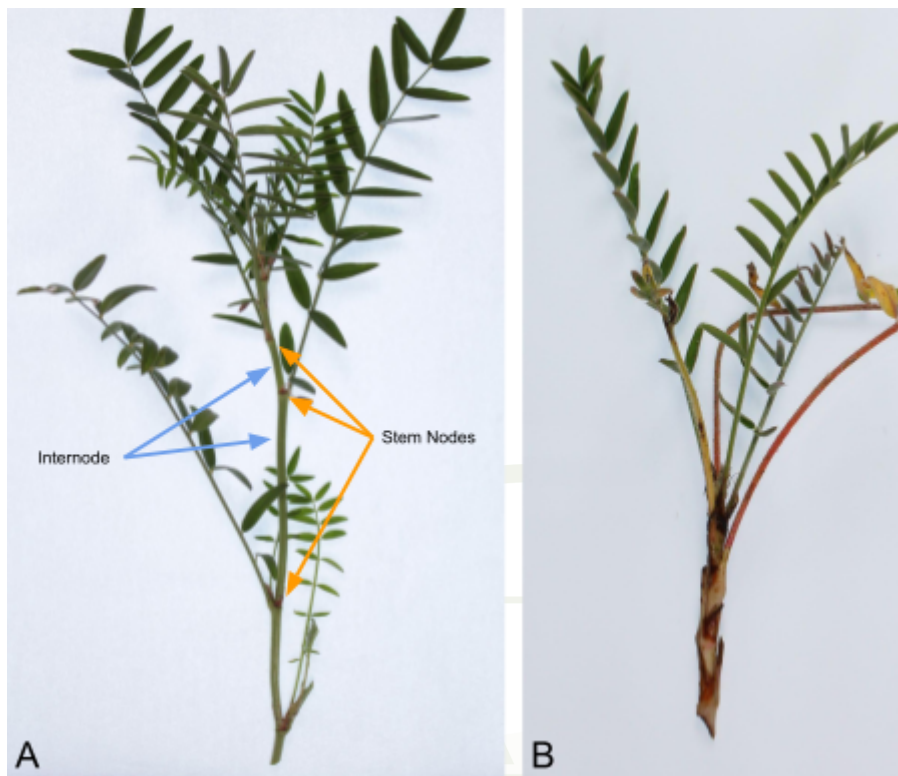


Figure 2. A) Sainfoin stem nodes, and internodes. B) A cutting taken from a crown bud.

DETAILED PROCEDURE

Planting Preparation

1. Moisten perlite in a container and fill as many pots as there are cuttings.
2. Fill a greenhouse tray with a layer of moistened perlite.
3. Make a 1% solution of Zeritol 2.0 in a spray bottle from the stock solution.

Propagation Technique

1. Select healthy tissue (reproductive stems or crown buds) that has 2 to 3 developing leaves (Figure 3).



Figure 3. Elongated Sainfoin stem.

2. Using a sharp, sterilized razor blade, make a 45 degree angle cut about 5 cm below the leaves for crown buds or 2 cm below a node for stem cuttings. Generally, the upper nodes on a stem will root quicker than lower nodes.



Figure 4. A) Sainfoin crown bud cutting. B) Sainfoin stem cuttings, each with 2 nodes present

3. Include 1 to 2 smaller leaves for crown buds and include at least 2 nodes if possible for stem cuttings (Figure 4). Remove any stem material or leaflets that have presence of disease (i.e. leaf spots, rust, etc.). Spray the whole cutting with the diluted Zeritol solution.

4. Dip the cutting in hormone (gel or powder) for a minimum of 1 minute and then place the cutting in the pot, ensuring the cutting is at least half way into the pot . 1 to 4 cuttings can be placed in the same pot. Label each pot with the plant genet name or plant ID.



Figure 5. Cuttings placed into a well drained pot with Perlite.

5. Sanitize the blade after each cutting with 70% ethyl alcohol to prevent transfer of disease.
6. Perlite pots can be placed in a tray with a layer of perlite underneath. If using a misting irrigation system, the tray should have drainage holes. If using water cans to irrigate, use solid trays to maintain a reservoir bed of moist perlite that will be wicked up into the pots.

Maintaining Cuttings

1. Place the trays on a greenhouse bench with a standard 18/6hr or 20/4hr light/dark cycle.
2. Mist the cutting every 1-2 hours with the automatic misting system. If unavailable, water the trays as the perlite begins to dry out. Avoid standing water in the trays.
3. Optional: Use a spray bottle of cloning solution to mist the cutting daily.
4. Optional: Use a plastic tray cover to maintain humidity if trays are drying too quickly. Be aware of possible molding issues if covers stay on too long.
5. Cuttings can be potted into general pots and standard potting soil mix once adventitious roots are present. Adventitious roots should be well developed 4 weeks after cutting.



Figure 6. Sainfoin crown bud cutting with adventitious roots after 4 weeks.

DATA STORAGE INSTRUCTIONS

1. NA

FLOWCHART

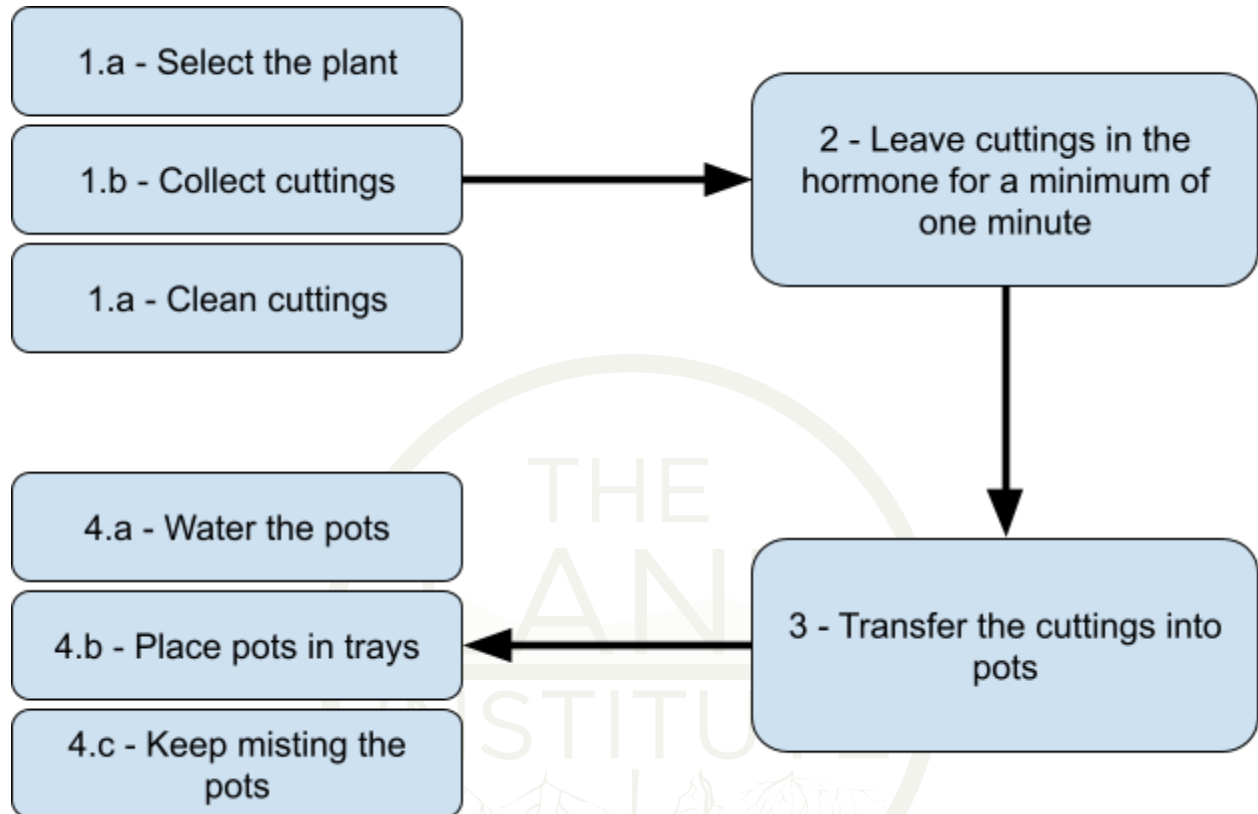


Figure 7. Flowchart for cloning Sainfoin.

TROUBLESHOOTING

- NA

NOTES

NA